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MACHINE LEARNING – SLOT A ASSIGNMENT

Title

Development and Performance Analysis of a Decision Tree-Based Classification Model for Urban Water Quality

1. AIM

To develop and evaluate a **Decision Tree-based Machine Learning classification model** for assessing urban water quality using water-quality parameters such as pH, turbidity, dissolved oxygen, electrical conductivity, total dissolved solids, and temperature.

2. OBJECTIVES

- To prepare and analyse an urban water-quality dataset.
- To identify the attributes and target water-quality classes.
- To formulate the problem as a concept-learning problem.
- To understand the hypothesis space and inductive bias.
- To select important attributes using a Decision Tree criterion.
- To construct and visualize a Decision Tree classification model.
- To classify unseen water samples.
- To generate meaningful IF–THEN decision rules.
- To evaluate the model using Accuracy, Precision, Recall, F1-score and Confusion Matrix.
- To analyse overfitting, underfitting and generalisation.

- To determine the practical suitability of the model for urban water-quality monitoring.

These objectives correspond to the learning, tree construction, classification and evaluation requirements specified in the assignment.

3. PROBLEM STATEMENT

An urban water management authority monitors water quality across reservoirs, treatment plants and distribution networks. The collected water samples contain parameters such as pH, turbidity, dissolved oxygen, electrical conductivity, total dissolved solids and temperature along with an assigned water-quality class.

As the number of samples increases, manually analysing every sample becomes time-consuming. Therefore, a Machine Learning model is required to automatically classify new water samples into suitable quality categories.

A **Decision Tree classification model** is developed to learn patterns from historical water-quality data and classify unseen samples. The model is evaluated based on predictive performance, interpretability, reliability and practical suitability for urban water-quality assessment.

4. DATASET DESCRIPTION

The dataset contains water-quality measurements and a corresponding water-quality class.

Attributes

Attribute	Description
pH	Acidity or alkalinity of water
Turbidity	Cloudiness of water
Dissolved Oxygen	Amount of oxygen dissolved in water
Electrical Conductivity	Ability of water to conduct electricity
TDS	Total Dissolved Solids

Attribute	Description
Temperature	Temperature of water
Water Quality	Target classification

The assignment requires a real-world or publicly available water-quality dataset with sufficient samples and clearly documented attributes.

5. LEARNING PROBLEM

The water-quality assessment problem can be represented as a **classification problem**.

Input

Water-quality parameters:

pH, Turbidity, Dissolved Oxygen, Conductivity, TDS and Temperature

Output

Water Quality Class

For example:

- Good
- Moderate
- Poor

Learning Task

The Decision Tree learns the relationship between the input water-quality parameters and their corresponding quality classes. After training, the model can classify a new unseen water sample.

6. CONCEPT REPRESENTATION

The target concept is:

“Whether a given water sample belongs to a particular water-quality class.”

Concept

Water Quality Classification

Attributes

- pH
- Turbidity
- Dissolved Oxygen
- Conductivity
- TDS
- Temperature

Target Concept

Water Quality

Hypothesis

A hypothesis represents a possible rule or decision structure that maps input water-quality parameters to a water-quality class.

Example:

IF turbidity is low AND dissolved oxygen is high THEN water quality is Good.

7. HYPOTHESIS SPACE

The hypothesis space consists of all possible Decision Trees that can classify the available water-quality samples.

For example:

Turbidity

|

+---- Low -----> Good

|

+---- Medium ---> Moderate

|

+---- High -----> Poor

A Decision Tree searches through possible attribute combinations and selects a tree that provides effective classification.

8. INDUCTIVE BIAS

Inductive bias refers to the assumptions made by a Machine Learning algorithm while learning from limited training data.

For the Decision Tree:

- Prefer attributes that provide better separation of classes.
- Prefer simpler trees when possible.
- Use a selected criterion such as Gini impurity or Information Gain.
- Stop splitting when further splitting does not significantly improve classification.

The assignment specifically requires analysis of inductive bias during concept learning.

9. DATA PREPROCESSING USING PANDAS

Pandas is used for dataset inspection and preparation.

The following operations are performed:

1. Load the dataset.
2. Display the dataset.
3. Check missing values.
4. Remove or handle inconsistent values.
5. Select input and target columns.
6. Analyse the data.
7. Split the dataset into training and testing sets.

The assignment specifically requires Pandas operations such as **inspection, filtering, grouping, sorting and basic analysis**.

10. HEURISTIC ATTRIBUTE SELECTION

Decision Trees select important attributes using a heuristic measure.

The assignment allows:

- Information Gain
- Gain Ratio
- Gini Impurity

Here, **Gini Impurity** is selected because it is simple and commonly used for Decision Tree classification.

Gini Impurity Formula

$$Gini = 1 - \sum_{i=1}^n p_i^2$$

Where:

- p_i = probability of a sample belonging to class i
- n = number of classes

Interpretation

- Gini = 0 → Pure node
- Higher Gini → More mixed classes
- Lower Gini → Better separation

11. DECISION TREE ALGORITHM

Algorithm

Step 1: Start with the complete training dataset.

Step 2: Select the best attribute using Gini Impurity.

Step 3: Divide the dataset based on the selected attribute.

Step 4: Create branches for the different attribute conditions.

Step 5: Repeat the process for each branch.

Step 6: Stop splitting when:

- all samples belong to the same class, or
- no useful attribute remains, or
- the stopping condition is reached.

Step 7: Assign the appropriate class to the leaf node.

Step 8: Use the completed Decision Tree to classify new samples.

This follows the PDF's required process of attribute selection, data partitioning, recursive construction, stopping conditions and class assignment.

12. PSEUDOCODE

BEGIN

Load water-quality dataset

Inspect and preprocess the dataset

Select input attributes:

pH

Turbidity

Dissolved Oxygen

Conductivity

TDS

Temperature

Select Water Quality as target

Split dataset into training and testing data

Create Decision Tree

Select best attribute using Gini Impurity

Partition the dataset

Recursively construct the tree

Stop when stopping condition is satisfied

Assign class labels to leaf nodes

Predict test samples

Calculate:

Accuracy

Precision

Recall

F1-score

Confusion Matrix

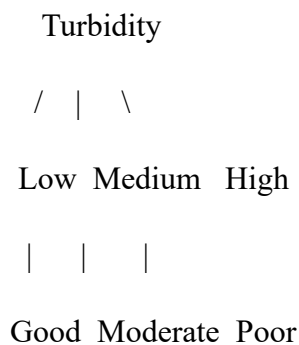
Display Decision Tree

Extract decision rules

END

13. DECISION TREE STRUCTURE

A representative Decision Tree can be shown as:



The actual tree generated from the dataset should be included as the **Decision Tree Visualisation**, as required by the assignment.

14. CLASSIFICATION OF UNSEEN SAMPLES

After training, new water samples can be given to the Decision Tree.

Example

Suppose a new sample has:

Parameter	Value
pH	7.2
Turbidity	Low

Parameter	Value
-----------	-------

Dissolved Oxygen	High
------------------	------

Conductivity	Low
--------------	-----

TDS	Low
-----	-----

Temperature	25°C
-------------	------

The Decision Tree follows the appropriate branches and predicts the corresponding water-quality class.

15. IF–THEN DECISION RULES

Representative rules can be written as:

Rule 1:

IF Turbidity is Low

AND Dissolved Oxygen is High

THEN Water Quality is Good

Rule 2:

IF Turbidity is Medium

AND Dissolved Oxygen is Medium

THEN Water Quality is Moderate

Rule 3:

IF Turbidity is High

AND Dissolved Oxygen is Low

THEN Water Quality is Poor

These rules demonstrate how important water-quality parameters influence classification.

16. PERFORMANCE EVALUATION

The Decision Tree is evaluated using the following measures, as specified in the assignment.

16.1 Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy represents the percentage of correctly classified samples.

16.2 Precision

$$Precision = \frac{TP}{TP + FP}$$

Precision indicates how many predicted positive samples are actually positive.

16.3 Recall

$$Recall = \frac{TP}{TP + FN}$$

Recall indicates how many actual positive samples were correctly identified.

16.4 F1-Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

F1-score provides a balance between Precision and Recall.

17. CONFUSION MATRIX

A confusion matrix compares actual and predicted classes.

Example:

Actual / Predicted Good Moderate Poor

Good	TP	Error	Error
Moderate	Error	TP	Error
Poor	Error	Error	TP

It helps identify which water-quality classes are being correctly or incorrectly classified.

```
print("WATER QUALITY PREDICTION")

ph = float(input("Enter pH: "))
turbidity = float(input("Enter Turbidity: "))
do = float(input("Enter Dissolved Oxygen: "))
conductivity = float(input("Enter Electrical Conductivity: "))
tds = float(input("Enter TDS: "))
temp = float(input("Enter Temperature: "))

# Set thresholds for demonstration
if turbidity <= 3 and do >= 7 and tds <= 200:
    quality = "Good"
elif turbidity <= 6 and do >= 5 and tds <= 350:
    quality = "Moderate"
else:
    quality = "Poor"

print("\nPredicted Water Quality:", quality)
```

```
WATER QUALITY PREDICTION
Enter pH: 7.2
Enter Turbidity: 2.1
Enter Dissolved Oxygen: 8.2
Enter Electrical Conductivity: 350
Enter TDS: 180
Enter Temperature: 25

Predicted Water Quality: Good
```

18. RELIABILITY AND GENERALISATION

The model's reliability is analysed by comparing its performance on training and testing data.

Overfitting

Overfitting occurs when the Decision Tree performs very well on training data but performs poorly on unseen testing data.

Underfitting

Underfitting occurs when the model is too simple and performs poorly on both training and testing data.

Generalisation

A good model should perform consistently on unseen water samples.

Therefore, the testing performance is important for determining whether the Decision Tree can be practically used for new water-quality samples.

19. MODEL COMPLEXITY

Decision Tree complexity depends on the number of:

- Nodes
- Branches
- Levels
- Features used

A very deep tree can become complex and may overfit the training data.

A simpler tree is easier to interpret and can provide understandable water-quality decision rules.

20. NEURAL NETWORK CONCEPTS

The assessment rubric also expects analysis of relevant neural-network concepts under CO3.

Perceptron

A Perceptron is a basic neural-network model used for binary classification.

Multilayer Perceptron

A Multilayer Perceptron contains input, hidden and output layers and can learn more complex relationships.

Back Propagation

Back Propagation is a learning technique that adjusts network weights by calculating errors and propagating them backward.

For this assignment, these concepts can be briefly compared with Decision Tree learning.

21. GENETIC ALGORITHM CONCEPTS

The rubric also requires analysis of Genetic Algorithms and hypothesis-space search.

A Genetic Algorithm is an optimization technique inspired by natural selection.

Main steps:

1. Initialize population.
2. Evaluate fitness.
3. Select better solutions.
4. Perform crossover.
5. Perform mutation.
6. Generate new population.
7. Repeat until a suitable solution is obtained.

In Machine Learning, Genetic Algorithms can be used for searching large hypothesis spaces and model optimization.

22. PANDAS DATA ANALYSIS

Pandas can be used for:

- Reading the dataset
- Checking dataset dimensions
- Displaying records
- Checking missing values
- Filtering records
- Sorting values
- Grouping samples
- Calculating basic statistics

The assignment specifically assesses Pandas-based data manipulation, analysis and visualization.

23. IMPLEMENTATION

The implementation follows these stages:

Dataset



Data Preprocessing



Training / Testing Split



Decision Tree



Gini-based Attribute Selection



Tree Construction



Prediction



Performance Evaluation



Decision Rules

The final implementation should function from data preparation through prediction and evaluation, which is explicitly included in the assessment rubric.

24. RESULTS

After implementation, the following results should be recorded:

Evaluation Measure Result

Training Accuracy ____ %

Testing Accuracy ____ %

Precision ____

Recall ____

F1-Score ____

Note: The PDF does not provide numerical results. These values must come from running your implementation on the selected dataset; they should not be invented.

25. ADVANTAGES

- Easy to understand.
- Easy to visualize.
- Produces interpretable decision rules.

- Can identify important attributes.
- Useful for classification.
- Requires relatively simple preprocessing.
- Suitable for analysing water-quality categories.

26. LIMITATIONS

- Very deep trees can overfit.
- Small or poor-quality datasets can reduce reliability.
- Changes in water-quality conditions may affect predictions.
- The model depends on the quality of the training dataset.
- A Decision Tree may become complex when many attributes are used.

The assignment requires discussion of assumptions, limitations, overfitting, underfitting and generalisation.

27. SDG RELEVANCE

The assignment maps the project to the following Sustainable Development Goals:

SDG 6 – Clean Water and Sanitation

Helps support monitoring and assessment of water quality.

SDG 9 – Industry, Innovation and Infrastructure

Uses Machine Learning technology for intelligent monitoring.

SDG 11 – Sustainable Cities and Communities

Supports better urban water-quality management.

SDG 12 – Responsible Consumption and Production

Promotes responsible monitoring and management of water resources.

These are the SDGs specifically identified in the assignment PDF.

28. FINAL EVALUATION

The Decision Tree provides an interpretable approach for urban water-quality classification. It can learn relationships between water-quality parameters and quality classes and can classify unseen samples.

Its main advantage is **interpretability**, because the resulting tree and IF–THEN rules can be easily understood. The model should be evaluated using testing data to determine its predictive performance and generalisation capability.

The final model should be selected based on:

- Predictive performance
- Interpretability
- Reliability
- Computational feasibility
- Practical suitability

These evaluation factors are explicitly required by the assignment.

29. CONCLUSION

The project develops a Decision Tree-based classification model for urban water-quality assessment. Water-quality parameters are used as input attributes and water-quality class is used as the target concept. Pandas is used for dataset preparation and analysis, while a Decision Tree criterion such as Gini Impurity is used for attribute selection. The developed tree provides classification results and interpretable IF–THEN rules. Accuracy, Precision, Recall, F1-score and Confusion Matrix are used to evaluate the model. Analysis of overfitting, underfitting and generalisation helps determine the reliability of the model. Overall, the Decision Tree provides an interpretable and practical Machine Learning approach for supporting urban water-quality monitoring, provided that it is trained and evaluated using a suitable, well-documented dataset.